RESUME

Alberto Boffi

Computer science engineer specializing in crafting scalable full-stack applications and and developing AI-driven solutions. Combining a deep mathematical background with an architectural foundation spanning data science, computer graphics, cybersecurity, advanced algorithms and several computing domains.

Milan, Italy | albertoboffi.1999@gmail.com | +39 3883542911 | LinkedIn: alberto-boffi | Github: albertoboffi | Website: albertoboffi.github.io

WORK EXPERIENCE

May 2024 - Present — Software Engineer, *Connect Reply* (Milan, Italy) - Full time.

- Key reference for a large web system for connected devices, serving 7+ countries and ∼500K users.
- Owner and developer of an upcoming application for the USA market, targeting 1M users.
- Engineering complex data exchange features across devices, UIs, and BE services, leveraging autonomous agents and LLM-driven solutions.
- Led the full SDLC, interfacing with global stakeholders, to ensure a reliable E2E delivery process.
- Applying data science methodologies to drive data-informed decision-making.

June 2017 - Sept. 2017 — Software Engineer, *Il Biancospino SCS* (Como, Italy) - Internship.

- E2E engineering and full-stack development of a web-based management system.

June 2016 - July 2016 — Software Developer, *Popcorn Creative Lab* (Como, Italy) - Internship.

- Developing institutional websites and web apps for tourism and business sectors.

EDUCATION

Sept. 2021 - April 2024 — Politecnico di Milano, *Master of Science*, Computer Science and Engineering.

Graduation grade: 110/110 cum laude.

Thesis title: VirtualRelativity: An Interactive Simulation of the Special Theory of Relativity in Virtual Reality.

Sept. 2018 - Sept. 2021 — Politecnico di Milano, *Bachelor of Science*, Ingegneria Informatica.

TECHNICAL STACK

FRONT-END: Angular, React, Redux, Svelte, TypeScript, JavaScript, jQuery, HTML5, CSS3, SCSS, Progressive Web Apps, Three.js, UI/UX Design.

BACK-END: Python, PHP, Node.js, Java, RESTful APIs, Event Streaming, GraphQL, gRPC, IAM, Microservices, MySQL, NoSQL, C#, Lua, Cybersecurity.

DEVOPS: Scrum, Kanban, Git, Github, CI/CD Pipelines, Docker, Sentry, New Relic, Azure DevOps, Jira, Confluence, SDLC coordination.

AI: TensorFlow, Scikit-learn (ML), Gymnasium (RL), Hugging Face, LangChain, Vector DBs, MCP, RAG, Prompt Engineering, Agentic Workflows, Deep Learning, LLM Integration, Google Colab.

DATA: Relational Design, ETL Pipelines, EDA, Analytics, A/B Testing, Data Security.

AUTOMATION: Claude Code, GitHub Copilot, Google AI Studio, Google Antigravity, Cursor, Google Jules, Google Opal.

EMBEDDED: VHDL, C, C++, IoT, SBC, Electrical Engineering, Hardware Architectures, Networking.

MEDIA: Figma, Canva, Unity (VR), Photo Editing, Vector Graphics, Video Editing, Google Stitch, Google Pomelli.

MATHEMATICS: Alghoritms & Data Structures, Operations Research, Formal Logic, Theoretical CS, Algebra, Discrete Mathematics, Physics, Calculus.

TECHNICAL WRITING

Progressive Web Apps: The Web's Answer to Native Applications. DEV Series [PDF version].

From One-Time-Pad to PKI: How Real-World Encryption Protects Our Data. DEV Series [PDF version].

Move Fast, Fail Fast, Judge Fair: Why Tech Needs Outcome-Based Responsibility. DEV Article [PDF version].

Boffi, A., Puppin, E., & Contran, M. (2024). VirtualRelativity: An Interactive Simulation of the Special Theory of Relativity in Virtual Reality. arXiv Preprint [Full thesis]

Boffi, A. (2021). An efficient way to manage ranges of data with Wise Red-Black Trees. arXiv Preprint.

PERSONAL PORTFOLIO HIGHLIGHTS

VirtualRelativity (2024) - VR ecosystem simulating the special theory of relativity. *ref*: Github, Website, Demo, Use Case, More

Human-Robot Interaction in Healthcare Scenarios (2022) - AI-driven framework for automatic configuration of virtual environments simulating HRI model in healthcare scenarios. *ref*: Technical Report

Friscool (2018) - Academic note-sharing web platform. *ref*: Demo

Fluo (2018) - Website composer developed as high school thesis project. Demo

[IN PROGRESS] *Moka PHP* - Lightweight PHP library for high-performance REST APIs. *ref*: Github, Website

[BETA] *Ludico* - Scalable social gaming PWA leveraging agent-automated SDLC over a regression-free framework. *ref*: Live Usage

[MVP] *Next-Gen Digital Pedagogy via Interactive STEM Technologies* - The first project in this initiative is an interactive 3D solar system simulator visualizing accurate planetary mechanics. *ref*: GitHub, Demo, Live Usage

LANGUAGES

Italian	Native.
English	Proficient (latest certificate TOEIC B2, 2021).